A Practical Market Making Algorithm for the IMC Algothon 2026 Challenge

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Abstract

We formalize a limit-order-book-based market making algorithm used for the IMC Algothon 2026 challenge. Our goal is to compute competitive bid-ask quotes while controlling adverse selection and inventory risk, and to convert those quotes into an execution policy that respects market microstructure constraints. We split the logic into three components: a pricing engine, an execution engine, and a position sizing engine.

The pricing engine decomposes quoting into (i) a theoretical price (fair value), (ii) a directional skew, and (iii) a half-spread. Unlike a classical microprice-centered approach, our theoretical price is a blended estimator combining the instantaneous midprice with a model-based expected real value estimate. This stabilizes quoting in noisy or thin books and allows the strategy to incorporate predictive information beyond top-of-book imbalance. The directional skew accounts for inventory risk and order book imbalance, while the half-spread depends on a base spread and realized volatility of the fair value estimator. We outline diagnostic analyses for signal validation, execution quality, and inventory/PnL behavior, and discuss potential extensions.

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1 Introduction

Our goal is to calculate bid-ask prices to maximize profits while controlling risks from adverse selection and inventory. We split the logic of the market making (MM) algorithm into three parts: pricing, execution, and position sizing. This modularity simplifies debugging and makes it easier to incorporate additional signals or refine execution behavior.

The pricing engine produces *model* (theoretical) bid-ask prices that are not yet rounded to legitimate ticks. The execution engine converts these prices into concrete order-management actions (join/improve/step-back, cancel/keep) while preserving queue priority in a price-time priority LOB. Lastly, the position sizing engine determines how much volume to post on each side while respecting inventory limits.

2 Pricing Engine

We focus on Level 1 LOB data; possible extensions using deeper book levels are discussed later. We exclude our own outstanding orders from the LOB when computing features, to

avoid self-induced feedback in signals.

Notation. For our resting quotes, denote O_i where $i \in \{\text{bid}, \text{bidvolume}, \text{ask}, \text{askvolume}\}$. For top-of-book data: V_{ask} and V_{bid} are TOB volumes, P_{ask} and P_{bid} are TOB prices, and τ is the tick size.

2.1 Theoretical Price / Fair Value (Blended Midprice + Expected Real Value)

Classical approaches often center quotes around a microprice computed from TOB prices and imbalance. In this strategy, the theoretical price is instead a *blend* of the instantaneous midprice and an *expected real value* estimate derived from predictive signals.

Own-order exclusion (TOB volumes). If we quote at the touch, we remove our own volume from the displayed TOB volume when computing imbalance-based features:

$$V^\star_{\text{bid}} = \begin{cases} V_{\text{bid}} - O_{\text{bidvolume}}, & \text{if } O_{\text{bid}} = P_{\text{bid}}, \\ V_{\text{bid}}, & \text{otherwise}, \end{cases} \qquad V^\star_{\text{ask}} = \begin{cases} V_{\text{ask}} - O_{\text{askvolume}}, & \text{if } O_{\text{ask}} = P_{\text{ask}}, \\ V_{\text{ask}}, & \text{otherwise}. \end{cases}$$

Midprice and spread.

$$M_t := \frac{1}{2}\big(P_{\text{bid},t} + P_{\text{ask},t}\big), \qquad S_t := P_{\text{ask},t} - P_{\text{bid},t}.$$

Expected real value. Let $\widehat{V}_t$ denote a model-based expected real value estimate. This could be produced by: (i) a regression/classifier on microstructure features, (ii) cross-asset/stat-arb signals, (iii) event/fundamental proxies (if available in the environment), or (iv) any predictive signal pipeline available in the competition.

Blended fair value. We define the theoretical price (fair value) as

$$F_t \;:=\; w\, M_t \;+\; (1-w)\, \widehat{V}_t, \qquad w \in [0,1]. \tag{1}$$

Intuition: M_t anchors to the immediate liquidity-based consensus, while $\widehat{V}_t$ injects directional information (alpha). The blend weight w controls responsiveness vs. stability and can be tuned per instrument.

Optional: microprice as an auxiliary signal (not the center). If desired, define a TOB microprice for diagnostics or as an additional feature:

$$I^{\text{raw}}_t := \begin{cases} \dfrac{V^\star_{\text{bid},t} - V^\star_{\text{ask},t}}{V^\star_{\text{bid},t} + V^\star_{\text{ask},t}}, & V^\star_{\text{bid},t} > 0,\ V^\star_{\text{ask},t} > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \mu_t := M_t + \frac{S_t}{2} I^{\text{raw}}_t.$$

In this write-up, quotes are centered around F_t (not μ_t).

2.2 Half-Spread

The half-spread is the distance from fair value to our bid/ask quotes. We keep it symmetric for simplicity and model it as a base term plus a volatility adjustment.

2.2.1 Base Spread

Let $s_0 \geq 0$ be a tunable base half-spread parameter (in price units or ticks, depending on implementation).

2.2.2 Realized Volatility (on Fair Value)

We use realized volatility computed on the fair value series $\{F_t\}$ and map it to ticks.

Volatility in ticks. Given a window $\{F_{t-w}, F_{t-w+1}, \ldots, F_t\}$ with window size w and tick size τ, define log-returns

$$r_k := \log\left(\frac{F_k}{F_{k-1}}\right), \qquad k = t-w+1, \ldots, t.$$

The per-step realized volatility is

$$\sigma_t^{(\text{per step})} := \sqrt{\frac{1}{w} \sum_{k=t-w+1}^{t} r_k^2}.$$

Mapped to ticks (using the current level F_t) gives

$$\nu_t := \sigma_t^{(\text{per step})} \frac{F_t}{\tau}.$$

Use $\nu_t = 0$ when $w < 1$.

2.3 Directional Skew

Directional skew shifts both bid and ask in the same direction to manage inventory and incorporate short-horizon directional information. We use order book imbalance and inventory risk.

2.3.1 Order Book Imbalance

Using the own-order-excluded volumes:

$$I_{\text{raw},t} := \begin{cases} \dfrac{V^\star_{\text{bid},t} - V^\star_{\text{ask},t}}{V^\star_{\text{bid},t} + V^\star_{\text{ask},t}}, & \text{if } V^\star_{\text{bid},t} > 0 \text{ and } V^\star_{\text{ask},t} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

We shape and scale into tick units (saturating nonlinearity):

$$\mathcal{I}_t := \tau \tanh\!\big(k\, I_{\text{raw},t}^3\big), \qquad k > 0.$$

2.3.2 Inventory Risk

Let current inventory be Q_t and symmetric position limit $Q_{\max} > 0$.

$$\bar{Q}_t := \min\{\, |Q_t|,\, Q_{\max} \,\}, \qquad x_t := \frac{\bar{Q}_t}{Q_{\max}} \in [0,1].$$

Define a smooth boosting function

$$b(x) := \frac{x^2}{x^2 + (1-x)^2}.$$

Inventory penalty in ticks:

$$\mathcal{R}_t^{\text{inv}} \;:=\; \tau\, b(x_t) \;=\; \tau\, \frac{x_t^2}{x_t^2 + (1-x_t)^2}\,.$$

Signed adjustment (push toward reducing inventory):

$$\widetilde{\mathcal{R}}_t^{\text{inv}} = -\,\operatorname{sgn}(Q_t)\,\mathcal{R}_t^{\text{inv}}, \quad \text{with } \operatorname{sgn}(0) := 0.$$

2.4 The Complete Pricing Formula

We clip the magnitude of components for stability.

Bounding (clipping) operator. For any scalar input ϕ and bounds $\phi_{\min} \le \phi_{\max}$,

$$\operatorname{clip}(\phi;\, \phi_{\min}, \phi_{\max}) = \min\big\{\phi_{\max},\, \max\{\phi_{\min},\, \phi\}\big\}.$$

Half-spread and skew.

$$h_t \;:=\; s_0 \;+\; \nu_t, \qquad \text{skew}_t \;:=\; \widetilde{\mathcal{R}}_t^{\text{inv}} \;+\; \mathcal{I}_t.$$

Optionally clip:

$$h_t \leftarrow \operatorname{clip}(h_t; h_{\min}, h_{\max}), \qquad \text{skew}_t \leftarrow \operatorname{clip}(\text{skew}_t; s_{\min}, s_{\max}).$$

Model bid/ask quotes (centered on blended fair value).

$$\begin{aligned} P_{\text{ask},t}^{\text{model}} &= F_t \;+\; h_t \;+\; \text{skew}_t, \\ P_{\text{bid},t}^{\text{model}} &= F_t \;-\; h_t \;+\; \text{skew}_t. \end{aligned} \tag{2}$$

3 Execution Engine

The execution engine converts model prices from the pricing engine into concrete order management actions while respecting tick size, spread constraints, and queue priority.

3.1 Objectives

1. **Stay competitive:** quote at or near the touch when the model supports it.
2. **Improve opportunistically:** step inside the spread when profitable and allowed by a spread gate.
3. **Avoid adverse selection:** step back when the model indicates negative edge or inventory pressure.
4. **Preserve queue priority:** avoid unnecessary cancellations.

3.2 Inputs

At each decision time t for side $s \in \{\text{bid}, \text{ask}\}$:

$$(P_{\text{bid}}, P_{\text{ask}}), \quad \tau, \quad \widehat{P}_s \in \{P_{\text{bid}}^{\text{model}}, P_{\text{ask}}^{\text{model}}\}, \quad O_s^{\text{px}} \text{ (optional)}.$$

3.3 Legal price and rounding

Prices are rounded to valid ticks and clipped to avoid crossing:

$$P_{\text{bid}}^{\text{legal}} = \min\left(\left\lfloor \frac{\widehat{P}_{\text{bid}}}{\tau} \right\rceil \tau,\ P_{\text{ask}} - \tau \right), \qquad P_{\text{ask}}^{\text{legal}} = \max\left(\left\lfloor \frac{\widehat{P}_{\text{ask}}}{\tau} \right\rceil \tau,\ P_{\text{bid}} + \tau \right).$$

3.4 Decision logic

Define signed tick distance to the touch:

$$\Delta_{\text{ticks}} = \begin{cases} \dfrac{P_{\text{bid}}^{\text{legal}} - P_{\text{bid}}}{\tau}, & \text{for bids,} \\ \dfrac{P_{\text{ask}} - P_{\text{ask}}^{\text{legal}}}{\tau}, & \text{for asks.} \end{cases}$$

With parameters j (join band), m (max improvement), b (max stepback), g (spread gate),

$$\begin{aligned} \textbf{Improve:} \quad & \Delta_{\text{ticks}} \geq j \ \text{ and } \ \tfrac{P_{\text{ask}} - P_{\text{bid}}}{\tau} \geq g, \\ \textbf{Join:} \quad & |\Delta_{\text{ticks}}| < j, \\ \textbf{Step back:} \quad & \Delta_{\text{ticks}} \leq -j. \end{aligned}$$

3.5 Queue preservation

Cancel/replace only if the new target price differs meaningfully and optional minimum quote lifetime has elapsed.

3.6 Outputs

$$(\text{cancel},\ \text{action},\ \text{price}),$$

where cancel $\in \{\texttt{cancel}, \texttt{nothing}\}$, action $\in \{\texttt{improve}, \texttt{join}, \texttt{stepback}, \texttt{nothing}\}$, and `price` is tick-aligned.

4 Position Sizing Engine

The sizing engine chooses posted volume on each side based on inventory and limits.

4.1 Inputs and Parameters

- Q_t: current inventory,
- $Q_{\max}$: position limit,
- $V_{\max}$: maximum order size,
- action_1: cancel decision from execution,
- O_s^{vol}: outstanding volume on side s.

4.2 Logic

$$\text{capacity} = \begin{cases} \max(0,\ Q_{\max} - Q_t), & \text{bid side}, \\ \max(0,\ Q_{\max} + Q_t), & \text{ask side}. \end{cases}$$

Subtract outstanding volume if not cancelling:

$$V_{\text{post}} = \max\big(0,\ \min(V_{\max},\ \text{capacity} - \mathbf{1}\{\text{no cancel}\}\, O_s^{\text{vol}})\big).$$

4.3 Implementation

```
def position_sizing_calculation(
        current_position, position_limit, max_post_volume,
        action1, side="bid", outstanding_position=None):
    outstanding_position = max(0, (outstanding_position or 0))
    if side == "bid":
        capacity = max(0, position_limit - current_position)
        if action1 != "cancel":
            capacity -= outstanding_position
    else:  # ask side
        capacity = max(0, position_limit + current_position)
        if action1 != "cancel":
            capacity -= outstanding_position
    volume = max(0, min(max_post_volume, capacity))
    return volume
```

5 Further Extensions

This section discusses practical extensions that improve robustness and allow the strategy to adapt to changing market conditions. We highlight (i) a multi-level microprice to reduce TOB noise sensitivity, (ii) markout-based diagnostics (and optional feedback control), and brief pointers to queue/fill and impact modeling, plus combining MM with statistical arbitrage.

5.1 Multi-Level Microprice

5.1.1 Motivation and Applications

Microstructure-based fair value estimators are central to high-frequency market making. The standard top-of-book (TOB) microprice reacts only to best bid/ask sizes and can be overly sensitive to fleeting orders, cancellations, or adversarial liquidity. Incorporating additional depth yields an estimator that is less noisy while remaining responsive to genuine shifts in supply and demand.

Such estimators are useful for:

- **Market making:** stabilizing the quote center and improving adverse-selection resilience.
- **Execution logic:** deciding whether to join, improve, or step back based on a more robust fair value.
- **Predictive signals:** deviations between depth-aware fair value and midprice often carry short-horizon information.

5.1.2 Classical Microprice and Its Limitations

Let B_1 and A_1 denote the best bid and ask prices with volumes V_1^b and V_1^a. The classical microprice is

$$\mathrm{MP}_{\mathrm{TOB}} = \frac{V_1^a B_1 + V_1^b A_1}{V_1^a + V_1^b}. \tag{3}$$

This construction weights each side's price by the *opposite* side's volume, reflecting the relative likelihood of the next trade occurring at that price. However, (3) ignores liquidity beyond the best level: large resting volume one or two ticks away can stabilize the price and should influence fair value.

5.1.3 Multi-Level, Distance-Weighted Microprice

Consider L bid levels (B_ℓ, V_ℓ^b) sorted $B_1 > B_2 > \cdots > B_L$ and L ask levels (A_ℓ, V_ℓ^a) sorted $A_1 < A_2 < \cdots < A_L$. Let tick size be $\tau > 0$. Define tick distances from the best quote:

$$d_\ell^b = \max\big\{\mathrm{round}\big((B_1 - B_\ell)/\tau\big),\, 0\big\}, \qquad d_\ell^a = \max\big\{\mathrm{round}\big((A_\ell - A_1)/\tau\big),\, 0\big\}. \tag{4}$$

Assign exponential distance weights with decay $\alpha > 0$:

$$w_\ell^b = e^{-\alpha d_\ell^b}, \qquad w_\ell^a = e^{-\alpha d_\ell^a}. \tag{5}$$

Define effective (discounted) volumes:

$$\widetilde{V}^b = \sum_{\ell=1}^{L} w_\ell^b V_\ell^b, \qquad \widetilde{V}^a = \sum_{\ell=1}^{L} w_\ell^a V_\ell^a, \tag{6}$$

and effective prices:

$$\widetilde{B} = \frac{\sum_{\ell=1}^{L} w_\ell^b B_\ell V_\ell^b}{\widetilde{V}^b}, \qquad \widetilde{A} = \frac{\sum_{\ell=1}^{L} w_\ell^a A_\ell V_\ell^a}{\widetilde{V}^a}. \tag{7}$$

The multi-level microprice is then

$$\mathrm{MP}_{\mathrm{multi}} = \frac{\widetilde{B}\widetilde{V}^a + \widetilde{A}\widetilde{V}^b}{\widetilde{V}^a + \widetilde{V}^b}. \tag{8}$$

Equation (8) reduces to (3) when $L = 1$ and $w_1^b = w_1^a = 1$. Optionally clamp $\mathrm{MP}_{\mathrm{multi}} \in [B_1, A_1]$ to guarantee the estimator lies inside the spread.

Interpretation. $\widetilde{V}^a/(\widetilde{V}^a + \widetilde{V}^b)$ can be interpreted as the probability of the next trade hitting the bid, while $\widetilde{V}^b/(\widetilde{V}^a + \widetilde{V}^b)$ corresponds to hitting the ask, with $\widetilde{A}$ and $\widetilde{B}$ incorporating nearby depth.

5.1.4 Worked Example

Consider $\tau = 0.1$ and spread $A_1 - B_1 = 0.2$ (two ticks):

$$\begin{aligned} B_1 &= 100.0, \quad V_1^b = 200, & B_2 &= 99.9, \quad V_2^b = 500, \\ A_1 &= 100.2, \quad V_1^a = 50, & A_2 &= 100.3, \quad V_2^a = 50. \end{aligned}$$

Let $\alpha = 1.0$. Then $d_1^b = 0$, $d_2^b = 1$, so $w_1^b = 1$ and $w_2^b = e^{-1} \approx 0.3679$ (similarly for the ask side). Effective volumes:

$$\widetilde{V}^b = 200 + 500e^{-1} \approx 383.94, \qquad \widetilde{V}^a = 50 + 50e^{-1} \approx 68.39.$$

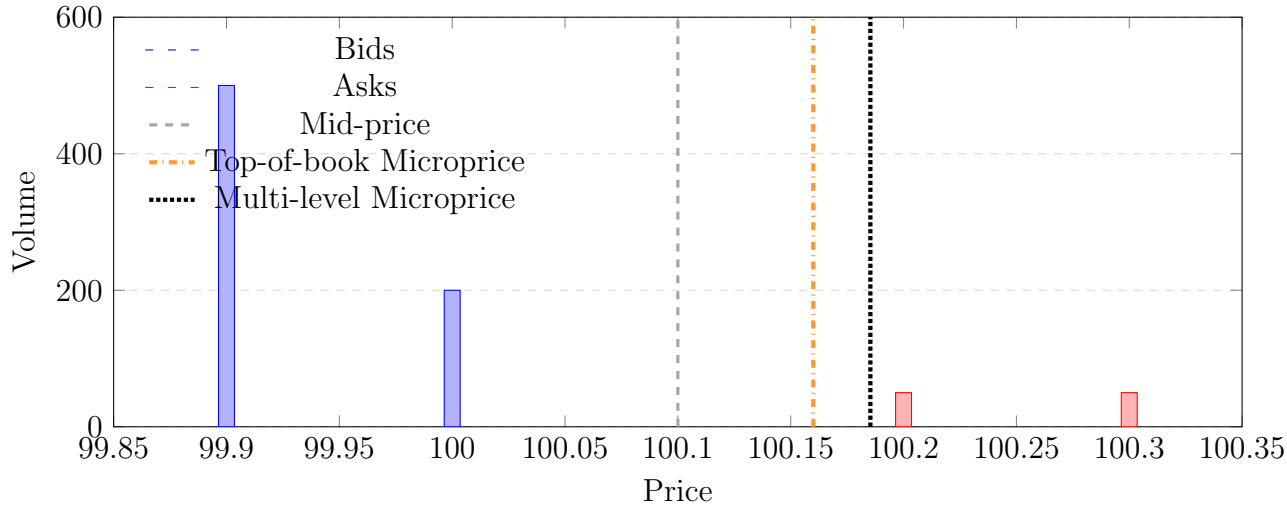
Effective prices:

$$\widetilde{B} \approx 99.9521, \qquad \widetilde{A} \approx 100.2269.$$

Substituting into (8) yields

$$\mathrm{MP}_{\mathrm{multi}} \approx 100.1853.$$

For comparison, the TOB microprice (3) gives $\mathrm{MP}_{\mathrm{TOB}} = 100.16$ and the midprice is 100.10. The multi-level microprice shifts closer to the ask, reflecting strong bid-side depth.



5.1.5 Parameter Considerations and Properties

Decay parameter α. Small α increases the influence of deeper levels; large α concentrates weight near the touch.

Truncation and stability. In practice, $L = 3$–5 levels are often sufficient because deeper levels contribute negligibly under exponential discounting. Guardrails include dropping zero-volume levels and reverting to midprice if $\widetilde{V}^a + \widetilde{V}^b$ is too small.

Scale invariance. Distances measured in ticks make the estimator robust to nominal price scale.

How to use with the blended fair value. A simple integration is to replace the midprice anchor in the blend:

$$F_t = w\,\mathrm{MP}_{\mathrm{multi},t} + (1-w)\,\widehat{V}_t,$$

which improves stability relative to TOB-only anchoring.

5.2 Multi-Level Order Book Imbalance

Analogous to the multi-level microprice, imbalance can be extended to incorporate depth. Using the same levels and weights as in (8), define discounted volumes:

$$\widetilde{V}^b = \sum_{\ell=1}^{L} w_\ell^b V_\ell^b, \qquad \widetilde{V}^a = \sum_{\ell=1}^{L} w_\ell^a V_\ell^a.$$

Then define the multi-level raw imbalance:

$$I_{\mathrm{raw}}^{\mathrm{multi}} := \begin{cases} \dfrac{\widetilde{V}^b - \widetilde{V}^a}{\widetilde{V}^b + \widetilde{V}^a}, & \widetilde{V}^b > 0,\ \widetilde{V}^a > 0, \\ 0, & \text{otherwise.} \end{cases}$$

As in the Level 1 design, shape and scale it into ticks:

$$\mathcal{I}^{\text{multi}} := \tau \, \tanh\!\big(k \, (I_{\text{raw}}^{\text{multi}})^3\big).$$

This reduces sensitivity to spoofing at the touch and uses nearby depth as confirmation.

5.3 Markout

In high-frequency market making, **markout analysis** evaluates execution quality and informational content by measuring how the market evolves after a fill. Markout can be used purely diagnostically (to detect adverse selection) or as a feedback signal to adjust spreads and aggressiveness.

5.3.1 Microprice Markout

Let μ_t denote a microstructure fair value proxy (TOB microprice or multi-level microprice), and suppose an order is filled at time t_0 at price P_{fill}. The k-step microprice markout is

$$M_{t_0,k}^{(\text{micro})} := \frac{\mu_{t_0+k} - \mu_{t_0}}{\tau}.$$

Interpretation: $M_{t_0,k}^{(\text{micro})} > 0$ suggests favorable drift after the fill; $M_{t_0,k}^{(\text{micro})} < 0$ indicates adverse selection.

Variant (relative to fill).

$$M_{t_0,k}^{\prime(\text{micro})} := \frac{\mu_{t_0+k} - P_{\text{fill}}}{\tau}.$$

This mixes price improvement and post-fill drift.

Practical usage. Persistently negative markouts imply quotes are being lifted just before unfavorable moves; common mitigations include increasing base spread, reducing improvements, or increasing inventory aversion.

5.3.2 Same-Side Markout

Suppose we are filled on side $s \in \{\text{bid}, \text{ask}\}$ at time t_0 at price P_{fill}. Let $P_t^{(s)}$ be the TOB price on the same side as the fill:

$$P_t^{(\text{bid})} = P_{\text{bid},t}, \qquad P_t^{(\text{ask})} = P_{\text{ask},t}.$$

Define trade sign

$$\eta_s := \begin{cases} +1, & s = \text{bid (we bought)}, \\ -1, & s = \text{ask (we sold)}. \end{cases}$$

Then the signed same-side markout is

$$M_{t_0,k}^{(\text{ss})} := \eta_s \, \frac{P_{t_0+k}^{(s)} - P_{\text{fill}}}{\tau}.$$

With this convention, $M_{t_0,k}^{(\text{ss})} > 0$ indicates a favorable outcome and $M_{t_0,k}^{(\text{ss})} < 0$ indicates adverse selection.

Averaging.

$$\overline{M}^{(\mathrm{ss})} = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} M^{(\mathrm{ss})}_{t_0,k}, \qquad \overline{M}^{(\mathrm{ss})}_{\mathrm{fills}} = \frac{1}{n} \sum_{i=1}^{n} M^{(\mathrm{ss})}_{t_{0,i},k}.$$

Relation to microprice markout. Microprice markout captures fair-value drift; same-side markout captures execution timing and adverse selection relative to the posted side of the book. Together they separate *directional drift* from *placement/queue effects*.

5.4 Fill Probabilities and Queue Modelling

A natural extension is to model fill probability as a function of distance to touch and queue position. A simple empirical model estimates

$$\mathbb{P}(\text{fill within } H \mid \delta) \approx f(\delta),$$

where δ is quote distance in ticks and H is a fixed horizon. In practice, $f(\delta)$ is well-approximated by a decreasing curve (often close to exponential). This can be used to optimize aggressiveness (join vs. improve) under adverse-selection constraints.

5.5 Price Impact Modelling

If trading larger sizes or reacting to alpha signals, incorporating impact reduces overtrading. A simple extension is to penalize expected impact in the quoting objective or widen spreads when short-horizon impact proxies increase (e.g., high trade rate, thinning depth, large imbalance changes).

5.6 Combination with Statistical Arbitrage

Market making can be combined with a stat-arb signal by injecting alpha into the theoretical price:

$$F_t = w\,(\text{microstructure anchor}) + (1-w)\,\widehat{V}_t,$$

or equivalently by adding an *alpha skew* component that shifts quotes in the predicted direction while respecting inventory and risk constraints. A practical approach is to: (i) keep the half-spread logic primarily volatility/liquidity driven, and (ii) add alpha into skew with clipping and regime gating to avoid overreaction in noisy periods.